



# Graphs in Machine Learning

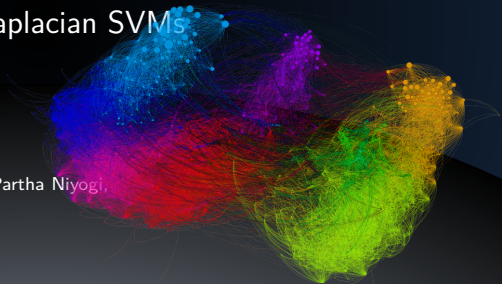
## SSL Manifold Regularization

Manifold Regularization and Laplacian SVMs

Michal Valko

*Isara Labs*

Partially based on material by: Mikhail Belkin, Partha Niyogi,  
Olivier Chapelle, Bernhard Schölkopf



# SSL with Graphs: Manifold Regularization

General (S)SL objective:

$$\min_f \sum_i^{n_l} V(\mathbf{x}_i, y_i, f(\mathbf{x}_i)) + \lambda \Omega(f)$$

Want to control  $f$ , also for the out-of-sample data, i.e.,  
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For general **kernels**:

$$\min_{f \in \mathcal{H}_{\mathcal{K}}} \sum_i^{n_l} V(\mathbf{x}_i, y_i, f(\mathbf{x}_i)) + \lambda_1 \|f\|_{\mathcal{K}}^2 + \lambda_2 \mathbf{f}^\top \mathbf{L} \mathbf{f}$$

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$$f^* = \arg \min_{f \in \mathcal{H}_{\mathcal{K}}} \sum_i^{n_l} V(\mathbf{x}_i, y_i, f) + \lambda_1 \|f\|_{\mathcal{K}}^2 + \lambda_2 \mathbf{f}^T \mathbf{L} \mathbf{f}$$

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$$f^* = \arg \min_{f \in \mathcal{H}_{\mathcal{K}}} \sum_i^{n_I} \max(0, 1 - yf(\mathbf{x})) + \gamma_A \|f\|_{\mathcal{K}}^2 + \gamma_I \mathbf{f}^\top \mathbf{L} \mathbf{f}$$



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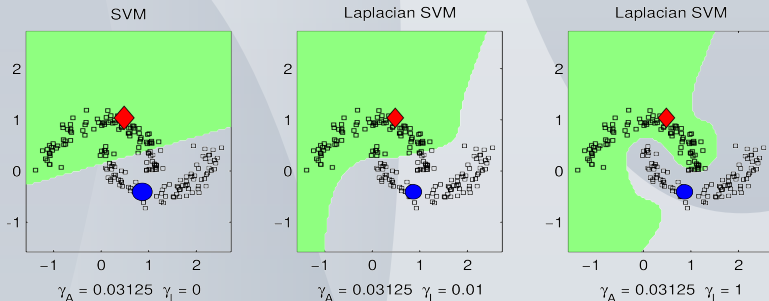
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Allows us to learn a function in **RKHS**, i.e., **RBf** kernels.

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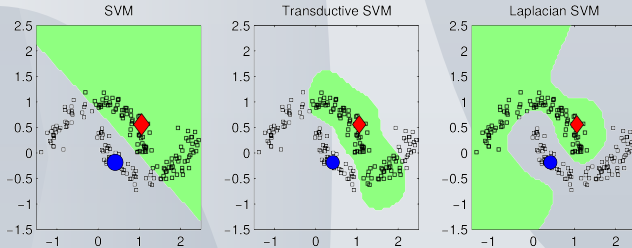
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Source: [belkin2006manifold](#) <empty citation>

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# Checkpoint 1

Semi-supervised learning with graphs:

$$\min_{\mathbf{f} \in \{\pm 1\}^{n_l + n_u}} (\infty) \sum_{i=1}^{n_l} w_{ij} (f(\mathbf{x}_i) - y_i)^2 + \lambda \sum_{i,j=1}^{n_l + n_u} (f(\mathbf{x}_i) - f(\mathbf{x}_j))^2$$

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Regularized harmonic Solution:

$$\mathbf{f}_u = (\mathbf{L}_{uu} + \gamma_g \mathbf{I})^{-1} (\mathbf{W}_{ul} \mathbf{f}_l)$$

## Checkpoint 2

Unconstrained regularization in general:

$$\mathbf{f}^{\star} = \min_{\mathbf{f} \in \mathbb{R}^N} (\mathbf{f} - \mathbf{y})^{\top} \mathbf{C} (\mathbf{f} - \mathbf{y}) + \mathbf{f}^{\top} \mathbf{Q} \mathbf{f}$$

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Out of sample extension: Laplacian SVMs

$$f^* = \arg \min_{f \in \mathcal{H}_{\mathcal{K}}} \sum_i^{n_I} \max(0, 1 - y f(\mathbf{x})) + \lambda_1 \|f\|_{\mathcal{K}}^2 + \lambda_2 \mathbf{f}^\top \mathbf{L} \mathbf{f}$$

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<https://misovalko.github.io/mva-ml-graphs.html>